Virtual Avatar Summary Report

Date: 3rd November 2026 **Timestamp:** 17:59:13 PM

Category: Technical

Learner Profile:

Name: Soumiya Ayyappan
Role: Graduate Apprentice | Domain: Graduate Apprentice
Tech Stack:

Initial Q&A Setup:

Q: Which technology or domain do you want to focus on?
A: databricks

Q: Is this for Concept Learning or a live issue?
A: Live Issue

Session Conversation:

[05:39 PM] Me:
Hello! Here is some background context about me for our session. My primary focus category for this session is technical. I am Soumiya Ayyappan, working as a Graduate Apprentice in the Graduate Apprentice domain. My tech stack includes . Here are some specific details about my project architecture: Regarding "Which technology or domain do you want to focus on?", my answer is "databricks". Regarding "Is this for Concept Learning or a live issue?", my answer is "Live Issue". Please use this information as context for my questions. Do not use a generic opening greeting. Instead, greet me as Phoebe, briefly summarize that you understand my background and focus area, and ask what we should focus on today!

[05:39 PM] PHOEBE:
Hello Soumiya, I understand you're a Graduate Apprentice focusing on Databricks for a live issue. What specific challenge or question do you want to

[05:39 PM] Me:
duplicates and then

[05:41 PM] Me:
So I'm not getting it, actually.

[05:41 PM] Me:
So just elaborate me on the silver layer part.

[05:42 PM] PHOEBE:
Certainly, Soumiya. The silver layer in a medallion architecture is primarily focused on refining and cleaning the data from the bronze layer. 1. Sequence of Execution: Typically, you start by analyzing the data for outliers, null values, and duplicates. Then, you clean the data by removing or correcting these issues before transforming it into structured tables. 2. Four Tables: Current Table: Contains the latest clean data. CDC (Change Data Capture) Table: Tracks changes and updates to the data over time. History Table: Maintains historical records of the data for auditing or analysis. Rejected Table: Stores records that did not meet validation criteria during the cleaning process. 3. Purpose: This process ensures the data is accurate, reliable, and ready for analysis in the gold layer. It helps in maintaining data integrity and provides a clear lineage of changes. What specific aspect of this process would you like to explore further?

[05:42 PM] Me:
with another tables. But there are so many now, so I'm not getting.

[05:45 PM] PHOEBE:
Certainly, Soumiya. Let's break this down: 1. Analyzing Null and Missing Values: This step is crucial to ensure data quality. Identifying these values helps you understand the completeness of your dataset and informs decisions on how to handle them—whether to fill, remove, or flag them. 2. Sales Events Processing: In your retail project, focusing on sales events is important because they drive business insights. Typecasting ensures data is in the correct format for analysis, while adding a flag column can help track specific conditions or issues related to the data. 3. CDC Purpose: Change Data Capture (CDC) is used to track changes in your data over time. By storing these changes in separate tables, you can maintain a history of updates, which is essential for auditing and understanding data evolution. 4. Multiple Tables: The reason for multiple tables is to organize data efficiently. Each table serves a specific purpose, allowing for easier analysis and management of different aspects of the dataset. Would you like to dive deeper into any specific step or concept?

[05:45 PM] Me:
also. Can you address me and also this, uh, CVC concept?

[05:47 PM] PHOEBE:
Certainly, Soumiya. Let's clarify these concepts: 1. CDC Procedure: Change Data Capture (CDC) captures changes made to your data over time. Before implementing CDC, you typically clean your data (removing nulls and duplicates) to ensure that only relevant changes are tracked. This helps maintain data integrity and provides a clear view of how data evolves. 2. Handling Outliers: Outliers are often not removed immediately because they can provide valuable insights. They may indicate unusual but significant events, and their treatment depends on the context of your analysis. It's essential to assess whether they are errors or important data points. 3. CVC Concept: I believe you meant "CDC" again. If you were referring to a different concept, please clarify. CDC helps in maintaining historical data and understanding trends over time, which is crucial for making informed business decisions. Would you like to discuss how to handle outliers or any specific CDC implementation details?

[05:47 PM] Me:
this.

[05:48 PM] PHOEBE:
Surely, Soumiya. Let's clarify these points: 1. Handling Duplicates vs. Outliers: In a single CSV scenario, filtering duplicates is straightforward. In a batch processing context, you may need to apply more complex logic to identify and handle duplicates across multiple files. 2. Removing Outliers: In machine learning, outliers can skew results and affect model performance. Techniques like Z-score or IQR can help identify outliers. However, the decision to remove them should be based on their impact on your analysis. 3. Batch Processing Complexity: Working with batch CSV files often involves additional steps like merging, aggregating, or transforming data to ensure consistency across the dataset. This can make the process seem more complex compared to handling a single file. Would you like to explore specific techniques for identifying outliers or discuss how to simplify batch processing?

[05:48 PM] Me:
which you understand? And while the same thing we are doing

[05:48 PM] Me:
in database. So kindly elaborate me on that.

[05:49 PM] Me:
Tell me that.

[05:50 PM] PHOEBE:
Certainly, Soumiya. In machine learning (ML) and database contexts, handling outliers, nulls, and duplicates is critical for data quality: 1. Outliers: In ML, outliers can be identified using statistical methods like Z-score or IQR. Depending on their significance, you may choose to remove, transform, or keep them for analysis. They can indicate anomalies or important patterns. 2. Nulls: Handling null values often involves imputation (filling in missing values), removal, or flagging them for further review. The approach depends on the data's context and the analysis goals. 3. Duplicates: Duplicates can skew results and should be identified and removed to ensure data integrity. Techniques like grouping or using unique identifiers help in this process. Each of these steps is essential for preparing your data for accurate analysis and model training. Would you like to dive deeper into any specific technique or aspect of this process?

[05:50 PM] Me:
me on that. According to my, this retail sales and, uh, in

[05:51 PM] PHOEBE:
Certainly, Soumiya. In the silver layer of your medallion architecture, you're focusing on refining and structuring your data for analysis. Here's how each component fits in: 1. Sales Events Table: This table contains cleaned and processed sales data. You perform operations like typecasting and null handling to ensure the data is accurate and usable for analysis. 2. Rejected Records Table: This table stores records that did not meet validation criteria during the cleaning process. It allows you to

track issues and understand why certain data points were excluded. 3. CDC Table: The Change Data Capture (CDC) table tracks changes in your data over time. It helps maintain a history of updates, which is crucial for understanding trends and changes in your retail sales

[05:51 PM] Me:
Tell me about the the the

[05:51 PM] Me:
all other things like basic

[05:52 PM] Me:
Pumping minds.

[05:52 PM] PHOEBE:
It seems

[05:52 PM] Me:
Yes. Tell me about optimize your

[05:52 PM] PHOEBE:
It seems there was a misunderstanding. Could you please clarify what specific aspect you would like to optimize? Are you referring to

[05:52 PM] Me:
I

[05:52 PM] PHOEBE:
It seems your message

[05:52 PM] Me:
no. I want to go

[05:52 PM] PHOEBE:
Understood, Soumiya.

[05:52 PM] Me:
no. No. I want

[05:52 PM] Me:
the basic things you're having. Me on that.

[05:53 PM] PHOEBE:
Certainly, Soumiya. In Databricks, optimizing data is crucial for performance. Here are the basics: 1. Optimize Z-Ordering: This technique improves query performance by co-locating related data in storage. It sorts data based on specified columns, making it faster to read when filtering. 2. Cluster Management: Databricks uses clusters to process data. Optimizing cluster size and configuration can enhance performance and reduce costs. 3. Liquid Clustering: This is a feature that allows

dynamic allocation of resources based on workload demands, ensuring efficient processing. Would you like to explore any of these concepts in more detail?

[05:53 PM] Me:
that

[05:53 PM] Me:
here and status updates, which we are doing that.

[05:53 PM] PHOEBE:
Surely, Soumiya. In the silver layer, regarding the rejected records: 1.

[05:53 PM] Me:
Tell me.

[05:54 PM] PHOEBE:
In the silver layer, the rejected records table captures data that failed validation checks during the cleaning process. This allows you to analyze why certain records were rejected, such as due to null values or incorrect formats. For the CDC (Change Data Capture), you typically merge the primary event table with the status updates table. This helps track changes in your data over time, ensuring that you have a complete view of all updates and modifications. Would

[05:54 PM] Me:
two. Can you

[05:54 PM] Me:
me on that?

[05:55 PM] PHOEBE:
Certainly, Soumiya. Merging the primary event table with the status updates table serves several purposes: 1. Comprehensive View: It provides a complete picture of each event's lifecycle, combining the original data with any updates or changes that have occurred. 2. Change Tracking: By merging these tables, you can track how specific events have evolved over time, which is crucial for understanding trends and making informed decisions. 3. Data Integrity: This process helps ensure that your analysis is based on the most current and accurate information, as it consolidates changes in one place. Would you like to explore how to implement this merging process in Databricks?

[05:55 PM] Me:
after that CDC in the... Saving as a current tape.

[05:56 PM] PHOEBE:
Certainly, Soumiya. To implement the merging process in Databricks, you would generally follow these steps: 1. Load Data: Load both the primary event table and the status updates table into your Databricks environment. 2. Merge Logic: Use a merge operation, typically with SQL or DataFrame API, to combine the two tables. You would match records based on a unique identifier (like an event

ID) and update fields from the status updates table where applicable. 3. Handle Duplicates and Conflicts: During the merge, decide how to handle any conflicts or duplicates. For example, you might want to keep the most recent status update or flag records that have discrepancies. 4. Store Results: After merging, save the resulting DataFrame as a new Delta table, often referred to as the "current table." This table will now contain the latest state of your data, incorporating both primary events and their status updates. 5. CDC Implementation: Finally, implement CDC by tracking changes in the current table over time,

[05:56 PM] Me:
Don't.

[05:56 PM] PHOEBE:
I understand, Soumiya. If you'd like to shift focus or have a different question, please let me know how I can

[05:56 PM] Me:
and rejected you completed. So why we are storing that in current?

[05:57 PM] PHOEBE:
Storing the merged data in the current table serves several key purposes: 1. Single Source of Truth: The current table acts as the most up-to-date representation of your data, integrating both primary events and status updates. 2. Efficient Access: Having a consolidated view allows for quicker access and analysis, making it easier to generate reports or perform queries on the latest data. 3. Facilitating CDC: By maintaining a current table, you can effectively implement Change Data Capture, tracking changes over time and ensuring that you have a historical record of data evolution. This helps in understanding trends and making informed business decisions. Would you like to explore any specific aspect of this process further?

[05:57 PM] Me:
so that I can do in a better way for my project.

[05:58 PM] PHOEBE:
Certainly, Soumiya. Let's summarize the silver layer process practically: 1. Sales Events Table: Start by cleaning your sales data. Ensure nulls and duplicates are handled. This is your foundational data. 2. Rejected Records Table: As you clean, any records that don't meet validation criteria are moved here. This helps you track issues and understand data quality. 3. CDC Table: This tracks changes over time. You will merge the primary event table with the status updates table to create a comprehensive view of each event's lifecycle. 4. Current Table: After merging, store the results in the current table. This table will represent the latest state of your data, integrating all updates and changes. By following these steps,

[05:58 PM] Me:
Okay. Thank you.